

Electric Arc Furnace Troubleshooting Guide

Overview

Troubleshooting guide for EAF-001.

Fault: Electrode Breakage

Symptom: Sudden loss of arc, current drops to zero. Cause: Scrap cave-in or mechanical stress.

Action: Retract electrode arms, remove broken pieces from the melt, install new electrode segment, tighten nipple joint to 2500 Nm.

Fault: Water Cooled Panel Leak

Symptom: High moisture in off-gas, drop in cooling water return pressure. Cause: Arc deflection or slag erosion. Action: Stop power immediately. Inspect panels visually. Replace leaking panel.

Fault: Hydraulic Roof Lift Failure

Symptom: Roof fails to lift or swing. Cause: Hydraulic pump failure or solenoid valve stuck. Action: Check pump pressure (>120 bar). Manually override solenoid valve to test.